

Olist SQL Analysis

Project Summary

Data Overview

The **Brazilian E-Commerce Public Dataset by Olist** contains real, anonymized marketplace transactions covering approximately **100,000 orders across Brazil between 2016 and 2018**. The dataset is organized into **9 relational tables** covering customers, orders, products, payments, logistics, and reviews, allowing for multi-dimensional e-commerce analysis. Product categories require Portuguese-to-English translation, while some logistics information may be incomplete for canceled or delayed orders. Overall, the dataset provides a strong foundation for analyzing revenue performance, customer behavior, product performance, and operational trends.

Business Insights

1. Revenue experienced strong growth but showed a major data anomaly

The monthly revenue analysis (Q2) shows strong growth from approximately **\$137K in January 2017 to more than \$1.15M by April 2018**, with revenue remaining above \$1M during much of 2018. However, revenue dropped dramatically to only **\$166 in September 2018**, which was also reflected in the **99.98% month-over-month decline identified in Q3**. This suggests that the September result may be related to incomplete reporting or a data-quality issue rather than normal business performance and should be investigated before being used for decision-making.

2. The customer base is heavily weighted toward one-time buyers

The customer analysis (Q7) identified **96,096 unique customers**, of which only **2,997 were repeat customers compared with 93,099 one-time buyers**. This indicates that the business has a significant opportunity to improve customer retention and convert first-time purchasers into repeat customers. Increasing repeat purchases could help strengthen customer lifetime value and provide a more sustainable source of revenue.

3. Revenue is led by Health & Beauty, while customer preferences vary by spending tier

The category analysis (Q4 and Q8) found that **Health & Beauty** was the top revenue-generating category, producing approximately **\$1.4M** and contributing **9.22% of total revenue**. The spending-tier analysis also shows that customer preferences differ: **High-tier customers** primarily spend on Watches & Gifts, Health & Beauty, and Sports & Leisure, while **Medium-tier customers** favor Bed, Bath & Table, Health & Beauty, and Sports & Leisure; **Low-tier customers** tend to purchase Telephony, Health & Beauty, and Computer Accessories. This suggests that category performance and customer spending behavior can be used to create more targeted product and marketing strategies.

Recommendations

1. Focus on customer retention and personalized marketing

Develop targeted retention campaigns to encourage the **93,099 one-time buyers** to make additional purchases, using follow-up promotions, loyalty incentives, and personalized product recommendations. Marketing can also be tailored to spending tiers by promoting categories that are most relevant to High-, Medium-, and Low-tier customers.

2. Strengthen category management while validating revenue data

Continue investing in high-performing categories such as **Health & Beauty**, while using the category and product rankings to optimize inventory, promotions, and cross-selling opportunities. At the same time, investigate the **September 2018 revenue anomaly** and validate the underlying data before using that period for performance comparisons or business decisions.